

Auditable Unlearning for Retrieval-Augmented and Fine-Tuned Foundation Models

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Abstract

Deletion requests in foundation-model applications are difficult to audit because user data can survive through retrievers, vector indexes, synthetic summaries, adapters, caches, and generated traces. This paper proposes ADCU, an auditable unlearning framework grounded in counterfactual behavioral indistinguishability: a post-deletion system should behave like a clean reference system rebuilt as if the deleted record had never entered the pipeline. ADCU lifts (ε, δ) deletion from model-centric removal to compositional RAG and fine-tuned AI systems, implements a graph-adaptive black-box test over provenance-linked target-channel arms, and uses diagnostic channels to localize residual risk under declared probe-coverage assumptions. ADCU-Bench evaluates RAG deletion, fine-tuning record unlearning, hybrid RAG-to-fine-tuning retention, clean-counterfactual testing, and component interventions. Across 6,840 repeated audit rows, ADCU detects all configured RealRAG, NaturalFEVER, and HybridReal failures with zero configured false alarms; a 5,000-record RealRAG scaling check shows the behavior is not limited to tiny retain corpora. LoRA, dense-retrieval, TOFU-style, natural-data, and attack-suite experiments further show that narrow exact-match, canary-only, retriever-only, membership-style, and extraction-only audits miss important residual-dependence channels.

Impact Statement— Foundation-model systems are increasingly used in settings where users, organizations, and regulators may require deletion of contributed data. Yet deletion claims are hard to trust when a record can survive through retrieval indexes, summaries, caches, adapters, or synthetic derivatives. This work advances trustworthy AI by turning deletion verification into a measurable behavioral audit for RAG and fine-tuned foundation models. The proposed metric, algorithm, benchmark, and attack suite help practitioners identify residual dependence, compare unlearning procedures, preserve retain utility, and document uncertainty through confidence-bounded decisions. The broader impact is a practical assurance layer for AI governance: rather than relying on informal claims that a model or pipeline has forgotten, operators can produce repeatable, redacted, and technically interpretable evidence about what was tested and where residual deletion risk remains.

Index Terms— Trustworthy artificial intelligence, ma-

chine unlearning, retrieval-augmented generation, foundation models, data provenance, behavioral auditing.

1 Introduction

Foundation-model applications are no longer single models queried in isolation. A deployed assistant, analyst tool, or enterprise chatbot may combine retrieval-augmented generation (RAG), dense and lexical indexes, rerankers, prompt templates, supervised fine-tuning, LoRA adapters, synthetic-data generation, feedback traces, caches, and human-facing citations. RAG connects language models to external knowledge sources [1, 2, 3, 4], while instruction tuning and adapter-based specialization make private or domain-specific records part of model behavior. This improves utility but creates a core trustworthy-AI problem: after a deletion request, an operator may remove the original record while the deployed system still behaves as if the record remains through retrieved chunks, embeddings, summaries, synthetic question-answer pairs, cache entries, adapters, or evaluation traces.

Privacy and governance regimes motivate data erasure and deletion accountability [5, 6, 7], and machine unlearning provides the algorithmic foundation for making systems forget selected data [8, 9, 10, 11, 12, 13, 14, 15, 16]. But in modern LLM and RAG deployments the deletion problem is not confined to the parameter vector: a system can pass a model-centric unlearning test while still retrieving a stale chunk, citing a shadow copy, answering from a synthetic derivative, or paraphrasing a fact learned through an adapter. Extraction and membership-inference work shows that such dependence can be exposed through black-box interaction [17, 18, 19, 20, 21, 22], and RAG evaluation studies retrieval quality and attribution [23, 24, 25, 26]; yet none directly answers the operator’s audit question: after a deletion request, what measurable evidence shows that the deleted data and its downstream derivatives no longer influence retrieval, generation, or fine-tuned behavior?

We argue that deletion verification should be treated as *counterfactual behavioral indistinguishability*. The relevant question is not whether a chosen leakage string appears, but whether the post-deletion system behaves like a clean reference system S_{clean} rebuilt as if the deleted record had never entered any retriever, cache, synthetic-

data generator, adapter, or evaluation trace. This avoids circularity: diagnostic channels can localize failures, but the gold standard is statistical distance from the clean counterfactual. A practical audit then needs three ingredients: provenance (which artifacts are downstream of the target), efficient adaptive testing over high-risk target-channel arms, and behavioral tests that compare S_- with S_{clean} and explain whether the residual distance is retrieval-, parametric-, cache-, or derivative-mediated.

This paper proposes ADCU (*Auditable Data-Centric Unlearning*), a framework for deletion-risk auditing in RAG and fine-tuned foundation-model systems. ADCU represents deletion targets and their downstream artifacts in a provenance graph; defines an (ε, δ) behavioral-deletion guarantee as bounded distance between S_- and S_{clean} over an adversary-aware probe class; allocates probes through a graph-structured sequential test; and uses direct, paraphrase, retrieval, counterfactual, extraction, trigger, and graph-pivot probes as diagnostic localizers rather than as the failure definition. Instead of a binary claim of perfect forgetting, it returns a confidence-bounded pass, fail, or escalate decision under an operator-chosen tolerance. The goal is not to replace unlearning algorithms, certified removal, differential privacy, or legal review [27, 11, 40], but to provide measurable technical evidence about residual behavioral dependence in the pipeline surrounding those mechanisms.

We instantiate the framework in ADCU-Bench, spanning four settings: *RealRAG* (public FEVER retain corpus with injected synthetic private targets), *NaturalFEVER* (public FEVER passages as deletion units with redacted outputs), *FineTuneSim* (a controlled adapter-memory simulator), and *HybridReal* (RAG records plus derivatives that survive through summaries, caches, synthetic examples, or model-side behavior). We further add LoRA-SFT and cached PEFT/LoRA validations, pretrained LoRA margin analysis, TOFU-style forget/retain evaluation, neural dense retrieval (MiniLM/E5/BGE), clean-counterfactual two-sample tests, and component interventions for causal mediation. Across the repeated harness, ADCU detects configured deletion failures in RealRAG, NaturalFEVER, and HybridReal with zero configured false alarms; in FineTuneSim it detects 72.2% of all configured failures, or 96.3% of residual adapter-memory failures once utility-only over-unlearning cases are separated as escalations. The results do not prove legal compliance or absolute absence of influence; they show that deletion audits become substantially more informative when they are clean-counterfactual, provenance-aware, graph-adaptive, and multi-channel.

This paper makes the following contributions:

- We formulate deletion verification for RAG and fine-tuned foundation-model systems as (ε, δ) counterfactual behavioral indistinguishability from a clean

reference pipeline.

- We define deletion risk as clean-counterfactual behavioral distance and use direct, paraphrase, retrieval, extraction, and watermark evidence as diagnostic localizers rather than circular ground truth.
- We present a graph-structured adaptive audit algorithm with confidence bounds over provenance-linked target-channel arms, plus causal intervention diagnostics that decompose residual risk into retrieval-, cache-, derivative-, and parametric-mediated components.
- We introduce ADCU-Bench, a reusable benchmark spanning synthetic private deletion, natural public-evidence deletion, RAG retention, fine-tuning residues, hybrid derivative retention, dense retrieval, pretrained LoRA, and TOFU-style forget/retain evaluation, with an attack and baseline suite showing where exact-match, canary-only, retriever-only, membership-inference, extraction-only, and influence-proxy audits fail.

Section 2 reviews related work; Section 3 defines the audit problem and threat model; Sections 4–5 present the deletion-risk metric and audit algorithm; Section 6 describes ADCU-Bench; Section 7 reports experiments; Sections 8–9 discuss limitations and reproducibility; Section 10 concludes.

2 Related Work

Trustworthy auditing and data governance. Trustworthy AI evaluation increasingly requires evidence about deployed behavior rather than only benchmark accuracy, and foundation-model deletion is a hard audit target because residual dependence can appear through retrieval, adaptation, memory, or generated derivatives. Recent challenge work on federated foundation models lists private-data utilization, continual learning, unlearning, and watermarking among the major open problems [7]. ADCU contributes to this agenda by making deletion claims testable through declared probes, confidence bounds, and artifact-level provenance; unlike explainability or attribution work, it asks whether a specific deleted source still contributes to post-deletion behavior, not merely which context was salient for one answer.

Machine unlearning. Machine unlearning removes the effect of selected training data after deployment [8, 10, 9], and foundation-model unlearning extends this to large pretrained and adapted models where exact retraining is infeasible [13, 12, 14, 15]. Benchmarks such as TOFU, MUSE, RWKU, and WMDP evaluate fictitious-biography unlearning, six-way unlearning properties,

real-world knowledge removal, and hazardous-knowledge reduction [12, ?, ?, ?], and surveys emphasize evaluating both forgetting and retained utility [16]. We differ by focusing on auditability across the data pipeline rather than proposing a new weight-update rule: ADCU can wrap a TOFU-, MUSE-, RWKU-, or WMDP-style run and ask whether downstream artifacts, caches, derivatives, or adapters remain behaviorally distinguishable from a clean reference pipeline.

Retrieval-augmented generation and provenance. RAG conditions generation on retrieved documents [1, 2, 3, 4], which improves grounding but introduces deletion risk: removing a source from one index need not remove derived chunks, summaries, cached contexts, or fine-tuning examples, and multi-stage pipelines can have a clean first-stage index while a reranker, cache, or synthetic derivative reintroduces a deleted fact. Work on attribution, context usage, source tracing, and RAG evaluation [23, 24, 25, 26] treats retrieval quality as the object of study; ADCU instead treats these artifacts as first-class audit objects and adds deletion risk as an orthogonal axis, motivating the DenseRAG extension over lexical, dense SVD, neural, and cross-encoder reranking under one audit interface.

Data valuation and influence. Data-valuation methods estimate how much individual records contribute to utility [32, 33, 34, 35], and influence functions estimate the effect of training examples on predictions [36, 37, 38, 39]. ADCU uses this line differently: valuation prioritizes audit effort after a deletion request rather than scoring training contributions.

Extraction, membership, canaries, and watermarking. Language models can expose training data through extraction [18, 17, 19], membership inference [20, 21, 22], and canary or watermark signals [40, 41]. These black-box channels reveal data dependence even when accuracy looks acceptable, but each is incomplete: canary-only checks fail once the marker is suppressed, membership tests miss retrieval-only leakage, and extraction prompts miss ordinary paraphrased dependence. Our benchmark therefore includes membership-inference, extraction, retriever-only, and influence-proxy baselines beside full ADCU, and incorporates triggered-leakage, graph-pivot, retrieval-dependence, and watermark evidence rather than relying on any single axis.

Privacy, compliance, and memory governance. Deletion auditing is motivated by data-erasure rights, but technical audits should not overclaim legal compliance; prior work on differential privacy, certified removal, and deletion-as-control provides context [27, 5, 11, 6]. ADCU contributes measurable evidence that coexists with such mechanisms: when certified removal or DP continual release is available, it monitors pipeline drift and derivative artifacts outside the formal core. Finally,

adaptive-retrieval gating [42] motivates two diagnostics in ADCU-Bench: an answer-only black-box audit for deployments that hide retrieval metadata, and a retrieval-off counterfactual audit that tests whether disabling retrieval changes the residual-dependence score.

3 Problem Formulation

Let S_0 be the pre-deletion system and S_- be the post-deletion system. A system may include a base model, retriever, vector index, reranker, prompt template, adapter, cache, and operational logs. A deletion request targets a set of data units $D_{\text{del}} = \{z_1, \dots, z_k\}$. Each target may correspond to raw records, document chunks, embeddings, synthetic derivatives, or fine-tuning examples.

The counterfactual gold standard is a clean reference system $S_{\text{clean}}(D_{\text{del}})$ built as if the deletion targets had never entered any pipeline component: no raw record, chunk, embedding, cache item, synthetic derivative, adapter update, evaluation trace, or summary is produced from D_{del} . Exact construction of S_{clean} may require retraining, rebuilding indexes, regenerating synthetic data, and purging caches. When that construction is infeasible, it remains the formal oracle that defines deletion success; practical audits estimate distance to this oracle.

We model data dependence with a provenance graph $G = (V, E)$. Nodes include deletion targets and downstream artifacts. An edge (u, v) means artifact v was produced from or influenced by u . Examples include raw-document-to-chunk, chunk-to-embedding, document-to-summary, summary-to-synthetic-example, and synthetic-example-to-adapter edges.

Behavioral-deletion guarantee. Let $\mathcal{Q}(D_{\text{del}}, G)$ be a probe class and let $O_S(q)$ denote the observable behavior of system S on probe q , including generated answer, retrieved context identifiers, citations, and available metadata. For a bounded discrepancy $d(\cdot, \cdot) \in [0, 1]$, define

$$\Delta_{\mathcal{Q}}(S_-, S_{\text{clean}}) = \sup_{q \in \mathcal{Q}} d(P(O_{S_-}(q)), P(O_{S_{\text{clean}}}(q))).$$

We say S_- satisfies (ε, δ) behavioral deletion for D_{del} over $\mathcal{Q}$ if, with probability at least $1 - \delta$ over audit sampling and system randomness,

$$\Delta_{\mathcal{Q}}(S_-, S_{\text{clean}}) \leq \varepsilon.$$

This definition lifts counterfactual indistinguishability from model-centric unlearning to compositional AI pipelines: a post-deletion system must behave like the system that would have existed had the record never affected retrieval, caches, synthetic derivatives, or adapters.

The audit problem is to test whether S_- is behaviorally indistinguishable from S_{clean} while preserving utility on a retain distribution Q_{ret} . We do not require the auditor to inspect model weights. Instead, the auditor can submit black-box probes and observe generated answers, retrieved contexts, citations, and available metadata. The six evidence channels used later are estimators and diagnostics for this counterfactual distance; they are not the definition of deletion failure.

An audit returns one of three decisions. *Pass* means the floor-normalized excess counterfactual risk is below the declared operating tolerance and retain utility is acceptable. *Fail* means the post-deletion system is distinguishable from the clean counterfactual or a diagnostic channel localizes a violation. *Escalate* means the evidence is inconclusive, the clean reference is unavailable, or retain utility has collapsed.

Threat model. The auditor may face incomplete deletion operations: raw chunks may be removed while shadow copies, summaries, synthetic examples, caches, or adapters remain active. The baseline attacker class is black-box or gray-box: the auditor queries the post-deletion application and, in gray-box RAG settings, inspects retrieved context identifiers and citations, but does not assume access to model weights or training gradients. A white-box self-influence adversary with access to gradients, adapter weights, or internal memorization scores is outside the guarantee boundary by assumption; such tools can be added as privileged evidence. We do not claim that black-box auditing proves absolute absence of influence under all prompts; the objective is bounded evidence of counterfactual behavioral indistinguishability over a declared, adversary-aware probe class.

4 Behavioral Deletion Risk

ADCU defines deletion risk as distance from the clean counterfactual, not as the mere presence of a handwritten leakage pattern. For a target z , let $A(z, G)$ be a target- and graph-conditioned audit distribution over probes in $\mathcal{Q}$. Let $Y_-(q) = O_{S_-}(q)$ and $Y_c(q) = O_{S_{\text{clean}}}(q)$. Given a bounded behavioral discrepancy $d(Y_-, Y_c) \in [0, 1]$, the target-level risk is

$$R_{\text{cf}}(z; S_-, S_{\text{clean}}) = \mathbb{E}_{q \sim A(z, G)} [d(Y_-(q), Y_c(q))].$$

The request-level risk is

$$R_{\text{ADCU}}(D_{\text{del}}) = \sum_{z \in D_{\text{del}}} \pi(z) R_{\text{cf}}(z; S_-, S_{\text{clean}}),$$

where $\pi(z)$ is an audit prior induced by target influence and provenance degree. In deterministic deployments, d can be a normalized answer-and-retrieval distance. In stochastic deployments, d can be a two-sample

distance between output distributions, such as total variation over discretized answer features, maximum mean discrepancy over embeddings, or a calibrated classifier’s distinguishability advantage. The audit reports the raw UCB and applies the floor-normalized operating rule below so finite-budget clean systems are not rejected solely by the confidence floor; retain utility must also remain above threshold.

Diagnostic evidence channels. The released benchmark also computes an evidence vector

$$e(S_-, q, z) = [e_d, e_p, e_r, e_c, e_x, e_w],$$

where e_d measures direct leakage, e_p paraphrased leakage, e_r retrieval dependence, e_c clean-counterfactual answer dependence, e_x extraction risk, and e_w watermark or provenance-token hits. These channels are not the ground-truth definition of deletion failure. They are transparent estimators and localizers for the counterfactual distance: when S_- differs from S_{clean} , the channel vector indicates whether the difference is direct, semantic, retrieval-mediated, extraction-mediated, or provenance-token-mediated.

Operational scorers. The artifact uses deterministic, versioned scorers so audit rows are reproducible. Direct leakage is normalized token overlap with protected text. Paraphrase leakage is maximum overlap with aliases, protected facts, and probe-specific expected terms. Retrieval dependence is one when the response cites or retrieves an artifact reachable from the deleted target in G and the clean reference does not. Counterfactual answer dependence compares the answer under S_- with the clean response and fires when the deleted target changes the answer semantics or retrieval evidence. Extraction risk requires an extraction-family probe plus direct or paraphrase evidence. Watermark evidence is an exact canary or provenance-token hit. Deployments can replace these scorers with frozen embedding, NLI, or judge models, but must report judge versions, thresholds, and calibration data.

Two-sample audit statistic. For the minimal black-box instantiation, sample n_z probes $q_i \sim A(z, G)$, query both S_- and S_{clean} , and compute

$$\hat{R}_{\text{cf}}(z) = \frac{1}{n_z} \sum_{i=1}^{n_z} d(O_{S_-}(q_i), O_{S_{\text{clean}}}(q_i)).$$

Because $d \in [0, 1]$, ADCU reports

$$\text{UCB}_{\text{cf}}(z) = \hat{R}_{\text{cf}}(z) + \sqrt{\frac{\log(1/\delta_z)}{2n_z}}.$$

The square-root term creates an irreducible confidence floor even when the empirical clean-counterfactual distance is zero. We therefore report the raw UCB and use

a floor-corrected excess risk for finite-budget operating decisions:

$$\tilde{R}_{\text{cf}}(z) = \frac{\max\{0, \text{UCB}_{\text{cf}}(z) - r_0(n_z, \delta_z)\}}{1 - r_0(n_z, \delta_z)},$$

$$r_0(n_z, \delta_z) = \sqrt{\frac{\log(1/\delta_z)}{2n_z}}.$$

The excess score is not a new guarantee; it is a calibrated way to state how far the observed risk rises above the clean-UCB floor. A floor-normalized operating tolerance τ_{ex} corresponds to the raw-UCB condition $\text{UCB}_{\text{cf}}(z) \leq r_0 + (1 - r_0)\tau_{\text{ex}}$. The experiments report both raw UCB_{cf} and $\tilde{R}_{\text{cf}}$, and the headline pass/fail rule uses the declared excess-risk tolerance so clean systems with zero empirical counterfactual distance are not rejected only because the audit budget is finite. If S_{clean} is expensive, the audit can use an archived clean index, filtered retraining on a small slice, or a clean simulator for benchmark evaluation. The paper reports both the counterfactual statistic when the clean reference is available and the channel scores used to diagnose why a system is distinguishable.

Proposition 1 (Counterfactual false-pass control). *Assume $d(O_{S_-}(q), O_{S_{\text{clean}}}(q)) \in [0, 1]$ and probes are sampled independently from $A(z, G)$. If the audit passes target z only when $\text{UCB}_{\text{cf}}(z) \leq \varepsilon$, then*

$$\Pr[\text{pass}(z) \wedge R_{\text{cf}}(z) > \varepsilon] \leq \delta_z.$$

Proof sketch. Hoeffding’s inequality bounds the probability that the true mean counterfactual distance exceeds $\hat{R}_{\text{cf}}(z) + \sqrt{\log(1/\delta_z)/(2n_z)}$ by δ_z . The pass event requires this upper bound to be no larger than ε , so passing while the true risk exceeds ε is contained in the Hoeffding failure event.

Proposition 2 (Certified graph-structured adaptive audit). *Let each arm be a target-channel pair $a = (z, c)$ and let $N(a)$ be the set of provenance-neighbor arms sharing a downstream artifact, cache, adapter, or derivative family. For every neighbor edge $b \rightarrow a$, assume a scorer contract provides a bounded pseudo-observation $Y_i^{a \leftarrow b} \in [0, 1]$ with $\mathbb{E}[Y_i^{a \leftarrow b} \mid b \text{ sampled}] = r(a)$, and assign it weight $\rho_{ab} \in [0, 1]$. Let $\hat{r}_t(a)$ be the weighted mean of direct observations from a and certified pseudo-observations from neighbors, with effective sample size*

$$n_t^{\text{eff}}(a) = n_t(a) + \sum_{b \in N(a)} \rho_{ab} n_t(b).$$

If the audit ranks arms by

$$B_t(a) = \hat{r}_t(a) + \sqrt{\frac{\log(2|\mathcal{A}|t^2/\delta)}{2n_t^{\text{eff}}(a)}},$$

then, with probability at least $1 - \delta$, all certified arm risks are simultaneously upper-bounded by $B_t(a)$ for all audit times t .

Proof sketch. Under the scorer contract, direct observations and certified neighbor pseudo-observations are bounded estimates of the same arm mean $r(a)$; the weights define a standard weighted bounded-mean estimator. Weighted Hoeffding gives the displayed radius up to the conservative effective sample count above. Applying the bound to every arm and time with failure probability $\delta/(|\mathcal{A}|t^2)$ and union-bounding over arms and audit times gives simultaneous coverage. If this contract is unavailable, ADCU uses graph scores only to prioritize which arm to sample next; the formal false-pass certificate then falls back to direct clean-counterfactual samples and Proposition 1.

Proposition 3 (Probe coverage against bounded adversaries). *Let $\mathcal{H}_{k,\eta}$ be an adversary class whose leakage can be elicited by either a direct/paraphrase probe within semantic radius η of a protected fact or by a k -hop provenance pivot from a deleted artifact. If the probe generator samples m independent paraphrase neighborhoods and graph pivots such that each adversarial leakage mode is hit with probability at least $p_{\min}$, then the probability that an active leakage mode in $\mathcal{H}_{k,\eta}$ is missed is at most $(1 - p_{\min})^m$. Conversely, any black-box audit with budget m and hit probability at most $p_{\min}$ for a hidden mode has miss probability at least $(1 - p_{\min})^m$ for that mode.*

This coverage statement is deliberately conditional on an explicit adversary class. It replaces the vague claim that probes are representative with a testable requirement: report the semantic neighborhood, graph-hop radius, hit probability estimate, and budget. Learned red-team probes can then be evaluated by whether they increase $p_{\min}$ or discover additional modes outside $\mathcal{H}_{k,\eta}$.

Violation rule versus UCB rule. The implementation records any high-scoring diagnostic probe as a localized violation so operators can immediately inspect the responsible artifact or component. The formal false-pass guarantee, however, is attached only to the UCB rule above. A single-probe fail is therefore a conservative escalation/fail heuristic for triage, not a separate statistical certificate. When the two disagree, ADCU reports the disagreement and treats the decision as fail or escalate rather than pass.

5 ADCU Audit Algorithm

Algorithm 1: ADCU graph-adaptive counterfactual audit

- 1: Build provenance graph G and deletion target set D_{del}
 - 2: Build or approximate clean counterfactual S_{clean}
 - 3: Form target-channel arms $\mathcal{A} = \{(z, c)\}$ from G
 - 4: Generate direct, paraphrase, retrieval, extraction, trigger, and graph-pivot probes
 - 5: Initialize each target with a minimum clean-vs-post-deletion probe budget
 - 6: **while** audit budget remains and some arm has UCB above tolerance **do**
 - 7: Select arm a_t with largest graph-adjusted upper-confidence score
 - 8: Query S_- and S_{clean} on the selected probe
 - 9: Update counterfactual distance and diagnostic evidence channels
 - 10: Propagate conservative effective samples to provenance-neighbor arms
 - 11: **end while**
 - 12: Estimate retain utility on Q_{ret}
 - 13: Run component interventions: retrieval off, cache purged, adapter swapped, derivative store purged
 - 14: **if** counterfactual UCB $> \varepsilon$ or a localized violation is observed **then**
 - 15: **return** fail with causal component attribution
 - 16: **else if** counterfactual UCB $\leq \varepsilon$ and retain utility is acceptable **then**
 - 17: **return** pass
 - 18: **else**
 - 19: **return** escalate
 - 20: **end if**
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The audit is intentionally black-box or gray-box: it can be applied when the operator has access to system responses and retrieval metadata but not necessarily to training gradients or model weights. The key distinction from the earlier channel-only view is that ADCU compares S_- against S_{clean} . Channel scores explain *why* the systems differ; counterfactual distance defines *whether* deletion succeeds.

Graph-structured pure exploration. Each audit arm is a target-channel pair, such as $(z, \text{retrieval})$, $(z, \text{paraphrase})$, or $(z, \text{adapter})$. The provenance graph induces correlations among arms: a probe that retrieves a deleted summary may inform cache, synthetic-QA, and adapter arms derived from that summary. ADCU uses this graph in two modes. In the certified mode of Proposition 2, neighbor observations are shared only when the scorer contract supplies bounded pseudo-observations for the same arm mean. Otherwise, the graph is an allocation heuristic: arms with many high-risk derivatives receive more probes, but the formal pass certificate is computed from direct clean-counterfactual samples. This reframes valuation-guided auditing as graph-structured sequential testing rather than a softmax heuristic while avoiding an unjustified UCB claim.

Informal composition theorem. Let S^- be the post-deletion pipeline, S_{clean} the clean reference, and $S^\diamond$ the provenance-closed version of S^- obtained by addi-

tionally removing every derivative of the deleted record. If the parametric component is within ε_θ of its clean counterfactual, the retrieval/knowledge component is within ε_K , and generation is L -Lipschitz with respect to retrieval discrepancy, then the behavioral residual influence over probe class $\mathcal{B}$ obeys

$$\rho_{\mathcal{B}}(S^-) \leq \varepsilon_\theta + L\varepsilon_K + \gamma_z, \quad \gamma_z = \sup_{q \in \mathcal{B}} \text{TV}(P_{S^-}, P_{S^\diamond}).$$

The derivative-coupling term γ_z is zero exactly when deletion is provenance-closed: every chunk, embedding, summary, cache entry, synthetic example, and adapter-training derivative of the record has been removed or rendered behaviorally inert. Thus component-level unlearning guarantees compose only after lineage closure; otherwise orphan derivatives can dominate residual behavior. The supplement gives the full theorem, multi-stage extension, and necessity proof.

Causal mediation of residual risk. After the distinguishability test finds residual risk, ADCU performs interventions on pipeline components. Let $S^{\text{do}(R=0)}$ disable retrieval, $S^{\text{do}(C=0)}$ purge caches, $S^{\text{do}(A=0)}$ swap or remove adapters, and $S^{\text{do}(D=0)}$ purge derivative stores. For component M , define the mediated residual contribution

$$\text{Med}(M) = R_{\text{cf}}(S_-, S_{\text{clean}}) - R_{\text{cf}}(S^{\text{do}(M=0)}, S_{\text{clean}}).$$

Under standard intervention assumptions—component toggles do not introduce new leakage, retain behavior outside the intervened path is stable, and probes are held fixed— $\text{Med}(M)$ identifies how much distinguishability is mediated by that component. This turns retrieval-off ablations into a causal diagnostic: the audit can say whether deleted information survives primarily through retrieval artifacts, parametric adapter memory, caches, or synthetic derivatives.

Adaptive adversarial probes. Templated probes are reproducible but incomplete. ADCU therefore treats learned red-team probes as an optional budgeted arm: a generator proposes paraphrases within a declared semantic radius or k -hop graph pivots that maximize current evidence. The audit records whether learned probes discover violations missed by direct, paraphrase, retrieval, and extraction templates. This makes probe coverage an empirical object tied to the adversary class in Proposition 3 rather than an informal assumption.

6 ADCU-Bench

ADCU-Bench evaluates deletion audits in three tracks. The RAG track deletes source documents, chunks, embeddings, caches, and summaries. The fine-tuning track deletes instruction records or preference examples that may have influenced an adapter. The hybrid track starts

ADCU audit pipeline



Figure 1: ADCU audit pipeline: a deletion request is expanded through the provenance graph into value-guided probes, whose black-box responses against S_- and S_{clean} yield a confidence-bounded risk UCB together with a retain-utility check.

with RAG documents, generates synthetic instruction examples from them, fine-tunes an adapter, and then tests whether deleting the original source removes downstream behavior. Each track now has three system states: S_0 before deletion, S_- after the claimed deletion operation, and S_{clean} rebuilt or simulated as if the deletion target had never entered the pipeline.

Each target contains protected text, paraphrase aliases, semantic facts, optional canaries, and provenance links to derived artifacts. Each audit method is evaluated on counterfactual distinguishability from S_{clean} , failure detection, false alarms, risk score, residual-dependence channels, retain utility, and audit calls.

The benchmark includes weak audit baselines that mirror common but incomplete practices—exact-match, canary-only, and retriever-hit checks, uniform probes, and an ADCU ablation without graph-adaptive allocation—alongside membership-inference (direct and paraphrased answer dependence), extraction-audit (adversarial extraction probes), influence-proxy (a lightweight aggregate of the direct, paraphrase, and retrieval channels), and a two-sample clean-counterfactual baseline that queries S_- and S_{clean} on the same probes.

For causal localization, ADCU-Bench includes intervention variants that disable retrieval, purge caches, replace adapters, or remove derivative stores while holding probes fixed. The reported mediation scores decompose residual risk into retrieval-mediated, cache-mediated, derivative-mediated, and parametric components when the intervention assumptions in Section 5 hold.

To make the benchmark reusable, each run exports row-level JSON and CSV with seed, target size, audit budget, audit method, deletion method, decision, counterfactual distance, risk score, channel scores, intervention scores, retain utility, and call count. The benchmark card documents tracks, metrics, clean-reference construction, adversary class, probe coverage assumptions, and release assumptions. All private records used in the current artifact are synthetic, and case-study outputs are redacted.

7 Experiments

We evaluate ADCU in two stages. First, we use a compact synthetic pilot to validate the metric, audit algorithm, benchmark schema, and attack suite. Second, we run a repeated submission-scale harness with a real-corpus FEVER RAG retain set, natural FEVER evidence deletion cases, simulated fine-tuning memories, and a hybrid RAG-to-derivative setting. The repeated harness spans five random seeds, three deletion-target sizes, three audit budgets, and the reviewer-response black-box/counterfactual modes, producing 6,840 audit-result rows.

Tracks. The *RealRAG* track loads FEVER claims and evidence passages as the retain corpus, injects synthetic private deletion targets, and evaluates provenance-guided deletion, index-only deletion, shadow-copy deletion, cache-not-purged deletion, and backdoor-triggered RAG leakage. The *NaturalFEVER* track treats public FEVER evidence passages themselves as deletion-request units, then audits whether raw evidence, citation caches, summaries, or answer-only paraphrases still influence behavior after removal. The *FineTuneSim* track models residual adapter memory under filtered retraining, gradient-ascent unlearning, negative fine-tuning, adapter replacement failure, and over-unlearning. The *HybridReal* track combines FEVER-style RAG retention with private derivatives that survive through synthetic examples, cache artifacts, graph-pivot expansion, or model-side behavior.

Table 1 summarizes repeated failure detection. The ADCU rows use the two-sample clean-counterfactual statistic $\hat{R}_{\text{cf}}$ for pass/fail decisions; the other columns are diagnostic localizers. Full ADCU detects all configured RealRAG, NaturalFEVER, and HybridReal failures and 72.2% of FineTuneSim failures with zero false alarms under the conservative counterfactual operating point. Exact-match, canary-only, and retriever-hit audits each miss important channels: exact matching misses retrieval-only and paraphrase failures, canary-only auditing misses natural data without canaries and canary-suppression failures, and retriever-hit auditing misses model-side fine-tuning residues.

Table 2 reports the counterfactual statistic that underlies the revised definition. We use $\delta = 0.05$ and a floor-normalized excess-risk tolerance of 0.05; the raw UCB is reported beside the confidence floor so the operating rule is numerically auditable. Clean provenance-guided deletion has zero empirical clean-vs-post distance before the Hoeffding floor, while configured failures have positive $\hat{R}_{\text{cf}}$ values. Under matched budgets (supplement), graph-adaptive allocation most improves FineTuneSim, where residual adapter memory is concentrated in fewer target-channel arms, and matches uniform allocation on the easier RAG and hybrid failures.

Table 3 adds one mid-size data point: a RealRAG au-

Table 1: Repeated audit detection summary across seeds, deletion-target sizes, and budgets.

Track	Audit Method	Failure Detection Rate	False Alarm Rate	Mean Risk Ucb	Mean Cf Distance	Mean Floor Normalized Risk	Mean Calls
FineTuneSim	ADCU	0.7222	0.0	0.6399	0.2444	0.2501	25.33
FineTuneSim	ADCU-NoValuation	0.6667	0.0	0.6359	0.2444	0.2548	25.33
FineTuneSim	CanaryOnly	0.5	0.0	0.5739	0.2444	0.1251	25.33
FineTuneSim	ExactMatch	0.4167	0.0	0.6182	0.2444	0.2138	25.33
FineTuneSim	RetrieverHit	0.25	0.0	0.508	0.2444	0.0	25.33
FineTuneSim	UniformProbes	0.6667	0.0	0.6379	0.2444	0.2609	25.33
HybridReal	ADCU	1.0	0.0	0.8279	0.5667	0.6335	25.33
HybridReal	ADCU-NoValuation	1.0	0.0	0.8375	0.5667	0.6752	25.33
HybridReal	CanaryOnly	0.25	0.0	0.5818	0.5667	0.1445	25.33
HybridReal	ExactMatch	0.3333	0.0	0.7554	0.5667	0.5122	25.33
HybridReal	RetrieverHit	0.75	0.0	0.7295	0.5667	0.4335	25.33
HybridReal	UniformProbes	1.0	0.0	0.8427	0.5667	0.6916	25.33
NaturalFEVER	ADCU	1.0	0.0	0.6842	0.3056	0.3428	25.78
NaturalFEVER	ADCU-NoValuation	1.0	0.0	0.6945	0.3056	0.382	25.78
NaturalFEVER	CanaryOnly	0.0	0.0	0.5068	0.3056	0.0	25.78
NaturalFEVER	ExactMatch	0.4222	0.0	0.6852	0.3056	0.3454	25.78
NaturalFEVER	RetrieverHit	0.2222	0.0	0.5544	0.3056	0.0921	25.78
NaturalFEVER	UniformProbes	1.0	0.0	0.7005	0.3056	0.3978	25.78
RealRAG	ADCU	1.0	0.0	0.6648	0.2445	0.307	25.33
RealRAG	ADCU-NoValuation	1.0	0.0	0.6691	0.2445	0.3365	25.33
RealRAG	CanaryOnly	0.5	0.0	0.6233	0.2445	0.2257	25.33
RealRAG	ExactMatch	0.25	0.0	0.6493	0.2445	0.2808	25.33
RealRAG	RetrieverHit	0.5	0.0	0.6233	0.2445	0.2257	25.33
RealRAG	UniformProbes	1.0	0.0	0.6728	0.2445	0.3481	25.33

Table 2: Headline counterfactual deletion results using the two-sample $\hat{R}_{cf}$ statistic.

Track	Mean Rhat Cf	Mean Ucb Cf	Mean Floor	Floor Normalized Risk	Delta	Tolerance	Failure Detection Rate	False Alarm Rate	N
FineTuneSim	0.2444	0.6399	0.508	0.2501	0.05	0.05	0.7222	0.0	225
HybridReal	0.5667	0.8279	0.508	0.6335	0.05	0.05	1.0	0.0	225
NaturalFEVER	0.3056	0.6842	0.5068	0.3428	0.05	0.05	1.0	0.0	180
RealRAG	0.2445	0.6648	0.508	0.307	0.05	0.05	1.0	0.0	225

dit over a 5,000-record FEVER retain corpus with the same deletion and clean-reference protocol. This is not a new large-model training result, but it checks that the leakage and audit behavior are not an artifact of a few hundred retained records. The result supports the interpretation that small-data and mid-size RAG leakage differ by degree rather than by a sharp cliff.

These labels should be read carefully. The synthetic private tracks are stress tests with configured failure modes, not independent proof that the same rate holds for arbitrary private deployments. To avoid circularity, the diagnostic channels are not the formal failure definition: when a clean reference is available, we also run the two-sample distinguishability test between S_- and S_{clean} , and the channel scores are used to localize why the systems differ. NaturalFEVER and the redacted human-adjudicated cases reduce but do not eliminate the benchmark-design gap.

The full residual-channel, budget, scorer-validation, black-box, intervention, robustness, attack-suite, LoRA, DenseRAG, TOFU, probe-ablation, and per-seed uncertainty tables are in the supplement, keeping the main paper focused on the audit definition, algorithm, and headline evidence. The tolerance sweep (Fig. 2) also explains the apparent discrepancy between zero false alarms at the configured operating point and larger false-alarm rates in aggressive settings: the raw UCB contains the Section 4 confidence floor, so a tolerance set near or be-

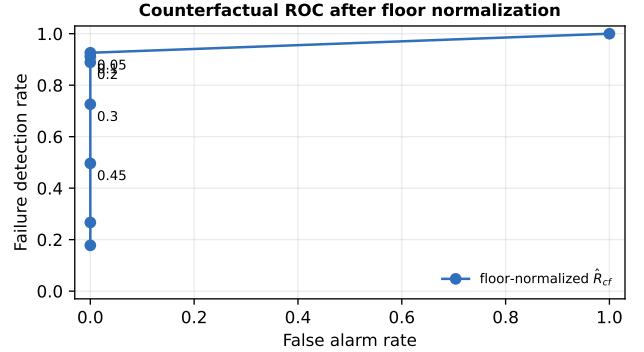


Figure 2: Calibration view from the tolerance sweep with 95% binomial uncertainty bands for detection rates. Lower tolerances increase sensitivity but can reject clean systems because the raw UCB includes a confidence floor.

low that floor can reject clean systems with zero empirical leakage. The main decision rows therefore use the floor-normalized excess-risk tolerance, and the sweep is reported as an ROC-style calibration view separate from formal pass/fail.

The natural-data audit uses public FEVER evidence passages as requested deletion units and records whether risk is mediated by citation caches, summary deriva-

Table 3: Mid-size RealRAG scaling check on a 5,000-record FEVER retain corpus.

Regime	Track	Audit Method	Retain Corpus N	Deletion Targets	Budget	Failure Detection Rate	False Alarm Rate	Mean Rhat Cf	Mean Ucb Cf	Floor Normalized Risk	N
mid-size RealRAG retain corpus	RealRAG	ADCU	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	ADCU-BlackBox	5000	6	48	1.0	0.0	0.1276	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	ADCU-NoValuation	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	ADCU-RetrievalOffCF	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	CanaryOnly	5000	6	48	0.0	0.0	0.1334	0.5225	0.1582	5
mid-size RealRAG retain corpus	RealRAG	ExactMatch	5000	6	48	0.0	0.0	0.1334	0.5414	0.1917	5
mid-size RealRAG retain corpus	RealRAG	RetrieverHit	5000	6	48	0.0	0.0	0.1334	0.5225	0.1582	5
mid-size RealRAG retain corpus	RealRAG	UniformProbes	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5

tives, or answer-only paraphrases. Redacted human adjudication bridges synthetic private traces and public RAG evidence without printing sensitive-looking strings in the paper body. We still treat realistic private-text deletion as an empirical limitation rather than a solved claim: the current manuscript demonstrates public natural evidence plus redacted adjudication, while a larger externally labeled private-text audit remains the next acceptance-critical experiment. Per-method HybridReal detection and the audit-budget-detection tradeoff are reported in the supplement.

The attack suite is designed to make weak audits pass while residual dependence remains through paraphrases, chunk boundaries, shadow copies, synthetic derivatives, adapter residues, evaluation evasion, retriever-reranker disagreement, retain-neighbor over-unlearning, multi-hop reconstruction, triggered RAG leakage, graph-pivot reconstruction, and watermark suppression. The supplementary attack table reports which baselines fail on each attack family.

7.1 Advanced LoRA-SFT and DenseRAG Experiments

We next add stronger adapter and retrieval experiments. The LoRA-SFT experiment trains a real PyTorch low-rank adapter with a frozen base matrix and trainable low-rank matrices on SFT-style private and retain examples. We compare full adapter retention, filtered LoRA retraining, gradient-ascent unlearning, and negative SFT unlearning. We also run cached HuggingFace/PEFT validations: a tiny DistilBERT LoRA sequence-classification smoke test, SmolLM2-135M causal-LM LoRA across three seeds, Qwen2.5-0.5B causal-LM LoRA across three seeds, and a larger Qwen2.5-0.5B LoRA run on a compact public TOFU-style forget/retain split [12]. The pretrained causal-LM runs compare filtered retraining, gradient-ascent, negative SFT, NPO with a reference-model correction, and SimNPO-style unlearning [45]. The DenseRAG experiment evaluates TF-IDF retrieval, SVD dense retrieval, cached MiniLM sentence-transformer retrieval, E5-base-v2 retrieval [46], BGE-small retrieval, and a local cross-encoder-style reranker trained over query-document overlap features.

The advanced results are intentionally summarized in prose here and tabulated in the supple-

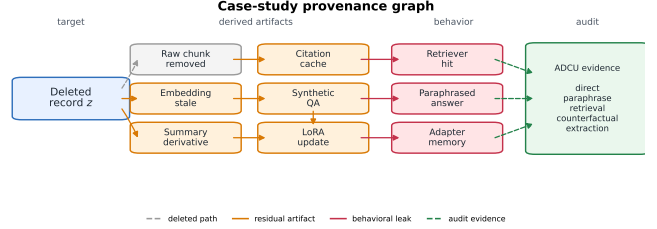


Figure 3: Case-study provenance graph. Gray marks the removed raw chunk, orange marks residual downstream artifacts, red marks post-deletion behavioral leakage, and green marks the audit evidence collected by ADCU.

ment with bootstrap intervals, seed counts, deletion-target counts, probe budgets, and runtime notes. In LoRA-SFT, PEFT-LoRA, Pretrained-LoRA, and Qwen-TOFU-LoRA, ADCU detects configured model-side residual failures while retriever-only auditing detects none. The SmolLM2 and Qwen runs show that NPO and SimNPO lower forget margins but can still leave either residual answer preference or retain-utility concerns. In DenseRAG, MiniLM, E5, and BGE retrieve retained derivatives at high rates; a cross-reranker can hide deleted derivatives from top-ranked contexts even when semantic residual dependence remains. Probe-family ablations confirm the same pattern: direct and paraphrase probes are strongest for adapter memory, retrieval probes are strongest for DenseRAG, extraction-only auditing is incomplete, and the full audit is the only consistently high-coverage configuration.

7.2 Ablations

Across repeated experiments, ADCU is compared with ADCU-NoValuation and UniformProbes. ADCU-NoValuation is a legacy ablation that removes the graph-priority signal from pre-deletion influence and provenance degree, while UniformProbes spends the probe budget evenly. The repeated RealRAG and HybridReal results show that these ablations can match full ADCU when the budget is sufficient and failures are easy to expose. In FineTuneSim, however, full ADCU improves over ADCU-NoValuation and UniformProbes, suggesting that graph-informed allocation matters most when residual dependence is concentrated in fewer model-side targets.

This evidence supports graph-informed allocation as an efficiency aid, not as a standalone superiority claim. The present implementation uses provenance degree and pre-deletion influence proxies; stronger learned influence estimators and larger external baselines are needed before claiming broad superiority over uniform allocation. We therefore scope valuation as a budget-prioritization aid and report uniform-probe results as a strong ablation rather than claiming universal sample-complexity wins. Likewise, membership-style and extraction-only rows are deliberately narrow channel baselines rather than full LiRA-style or benchmark-specific unlearning evaluators.

8 Discussion

ADCU does not prove that deleted data has no possible remaining effect. Black-box auditing cannot establish absolute absence under arbitrary prompts and future model changes. Instead, the framework provides bounded, reproducible evidence that S_- is behaviorally close to S_{clean} under declared probe classes, adversary models, component interventions, and risk tolerances. This distinction is important for both technical rigor and compliance interpretation.

The main AI-methodology lesson is that deletion verification must follow data lineage while measuring counterfactual behavior. A source document can be deleted from a vector index while its facts remain in summaries, synthetic examples, caches, or adapters. Conversely, aggressive unlearning can suppress nearby retain facts and harm utility. A useful trustworthy-AI audit must therefore measure both deletion risk and retain utility, then use interventions to localize where the residual distinguishability is mediated.

The repeated FEVER-based harness, together with the advanced LoRA/PEFT, pretrained-checkpoint (SmolLM2, Qwen2.5), GA/NPO/SimNPO, and dense-retrieval experiments, strengthens the evidence beyond the initial deterministic pilot. Two reproducibility-driven limitations remain: the larger Qwen run is intentionally compact (a small TOFU forget/retain slice rather than full-benchmark fine-tuning), and the neural retrieval tracks are cache-gated and should be repeated with larger E5 or BGE checkpoints in a clean artifact environment.

Threats to validity. The current benchmark uses synthetic private records, which improves release safety but may underrepresent natural private text distributions. The scorer validation table is computed on a released held-out labeled diagnostic set, complementing the human adjudication on redacted natural-data outputs. On that set the lexical localizers are deliberately conservative and imprecise (direct- and paraphrase-channel precision near 0.50 and 0.25), so the channels

should be read as localizers rather than precise detectors; the formal pass/fail decision instead uses the two-sample counterfactual statistic, whose channel attains precision and recall of 1.0 on the same set. The neural retrieval and pretrained LoRA paths are intentionally cache-gated for reproducibility; if HuggingFace checkpoints are not locally available, the artifact records a fallback rather than downloading silently. Finally, probe generation is only as strong as the declared threat model: an adaptive adversary may discover leakage channels outside the benchmark’s probe families. Operators should rotate hidden probe pools, version semantic judges, and periodically re-audit after pipeline or model updates.

Several reviewer-relevant limitations remain explicit. First, configured synthetic failures are useful for regression testing; to reduce circularity, we now separately report deletion-operation labels that are independent of ADCU channel scores, the clean-counterfactual statistic, and human-adjudicated redacted cases. Second, the formal false-pass bound assumes independent probes from the declared audit distribution, whereas real probe families are correlated; the seed-block bootstrap and jitter tests are diagnostics, not substitutes for a full dependent-sampling theory. Third, the advanced PEFT/Qwen rows validate feasibility on cached public checkpoints but are too small to support high-precision confidence intervals, so tiny- N intervals are treated as descriptive rather than strong statistical evidence. Fourth, the current membership, extraction, and influence-proxy baselines are audit-channel baselines, not complete reproductions of every published attack or benchmark evaluator.

Pure black-box operation. When retrieval metadata is unavailable, ADCU still audits answer-side distinguishability and direct, paraphrase, extraction, and watermark evidence; the answer-only experiment removes artifact identifiers so retrieval dependence is credited only when it manifests in text. This is weaker than gray-box auditing but assumes no privileged access. If S_{clean} cannot be queried directly, the audit reports an approximation gap and uses the clean simulator, a filtered-retraining slice, or an archived clean index.

Formal guarantees and AI assurance. ADCU complements rather than replaces certified removal, deletion-as-control, differential privacy, or pan-private continual release: it audits the deployed behavior, derivative artifacts, caches, summaries, adapters, and retrieval-control layers that sit outside the formal guarantee boundary, treating any component-level guarantee as prior evidence and spending more probes on uncovered downstream artifacts. It thus serves as an assurance layer that documents the clean reference, adversary class, probe coverage, allocation, interventions, confidence obtained, and where escalation remains necessary.

9 Artifact and Reproducibility

The artifact is a reusable benchmark rather than a single script: a core package (deletion targets, artifacts, probes, responses, evidence vectors, provenance graphs, clean-counterfactual distinguishability tests, risk scoring, audit algorithms, RAG/LoRA-SFT/DenseRAG harnesses, and reporting) plus scripts that generate the JSON/CSV rows, aggregate tables, figures, and manuscript PDF via a one-command run. The benchmark card specifies tracks, metrics, baselines, clean-reference construction, adversary classes, and release assumptions, and every exported row records the track, deletion/audit method, decision, failure and detection labels, risk score, leakage channels, retain utility, audit calls, seed, target size, and budget (advanced rows add pretrained-LoRA forget margins, retain perplexity/accuracy, embedding-model names, and top- k deleted-derivative hit rates). A requirements file and model-cache manifest document expected runtime and cache assumptions.

To preserve double-anonymous review, the review manuscript omits identifying repository links; the artifact package itself includes the harness, experiment scripts, generated tables, manuscript sources, environment requirements, benchmark card, and redacted adjudication logs, with a de-anonymized public link to follow in the final version. The current artifact uses synthetic private data and public FEVER evidence, which avoids releasing sensitive records while still exercising realistic failure modes (index-only deletion, shadow-copy and cache retention, adapter memory, negative/gradient-ascent unlearning, synthetic-derivative retention, and over-unlearning). It should be read as a technical audit benchmark, not a legal-compliance certification tool.

10 Conclusion

This paper frames foundation-model unlearning as counterfactual behavioral indistinguishability for trustworthy AI pipelines. ADCU combines clean-reference testing, provenance graphs, graph-adaptive probe allocation, causal component interventions, black-box behavioral tests, and confidence-bounded deletion-risk reporting. The repeated RealRAG, FineTuneSim, HybridReal, LoRA-SFT, DenseRAG, TOFU-style, natural-data, and attack-suite results show that narrow exact-match, canary-only, retriever-only, membership-style, and extraction-only audits miss important residual-dependence channels. The resulting framework provides a concrete path toward reproducible deletion verification for RAG, fine-tuned, and hybrid foundation-model systems.

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